Extending the lifespan of Noble Hill Landfill

***ANH VI** · Drury University · Instructor Dr. Leinaweaver*
***M.Arch** '25*
***Environment and Sustainability** Minor*
***Design Solutions to Environmental Problems** Certificate*

Introduction

Among 19 landfills in Missouri, Noble Hill Sanitary Landfill has been managing solid waste for Greene county and four other nearby counties since 1975. As part of the Integrated Solid Waste Management System, the landfill operates several other programs, including recycling centers, education, market development, and a yard waste sale center.

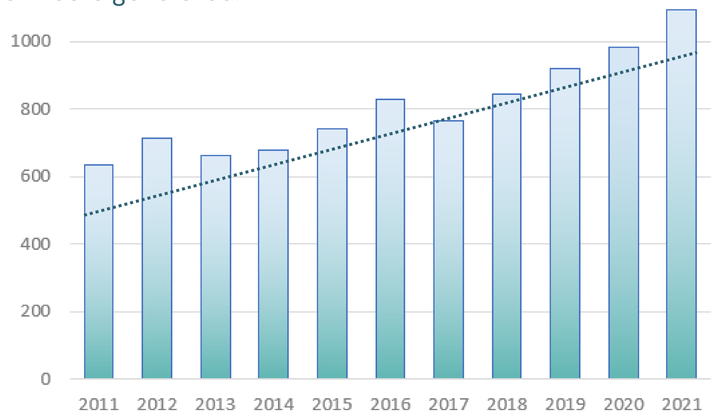
The City of Springfield's Department of Environmental Services is the local government agency that operate and oversee the program, while working with state and federal government to execute their goals to solid waste management.

CITY OF Springfield ENVIRONMENTAL SERVICES

Challenge:

The most critical resource that a landfill offers is space, and this resource is not unlimited. Even with the recent expansion, Noble Hill landfill is predicted to reach capacity and be closed down in 2112. In addition, waste production in Springfield has increased over the last decade, exacerbating the landfill space shortage.

The chart below shows a linear increase in the daily amount of waste generated:



Daily Tonnage received at the Landfill (tons/day)

Despite our landfill's spatial shortage, 70% materials in landfill are recyclable. Besides construction waste, the majority of recyclable materials are paper and food, which are generated by consumers. Effective recycling has been an ongoing challenge among individuals, households, and institutions.

26 %

Paper Products

25 %

Construction Waste

12.1 %

Food Waste

Source: *Tipping the scales: Springfield landfill filling up faster each year. Springfield Daily Citizen.*

POLICY DESIGNS

Command & Control Regulation

At present, waste collection and recyclable materials pick-up are treated as separate services, with residents often opting for the more economical option. The city government could mandate that households sign up for curbside recycling services.

Pro:
Immediate results.
Target all stakeholders.

Cons:
Inflexible to adjustments.
Do not accommodate poor households.

Green Taxes

City government can charge households based on their waste generation (per ton) or limit households to a certain number of permits for trash collection instead of monthly billings.

Pro:
Provide stakeholders with financial incentives.

Cons:
Challenging to set an appropriate tax rate.

Green Subsidies

The city can create a competition to promote recycling. Households with highest percentage of recycle to the overall waste generated will receive a grant or prize.

Pro:
Encourage innovative solutions from any group among the community.

Cons:
Where does the money come from?

Adaptive Governance

The city could host public meetings to gather feedback on an agreement aimed at reducing waste. The primary objective would be to raise awareness among the community about the issues at the Noble Hill Landfill.

Pro:
Encourage greater self-awareness and promote opportunities for taking compensatory actions.

Cons:
Challenge to acquire equal rights among different groups.